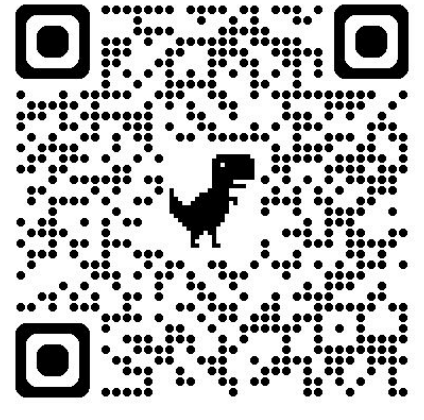


Query Circuits: Explaining How Language Models Answer User Prompts

Paper

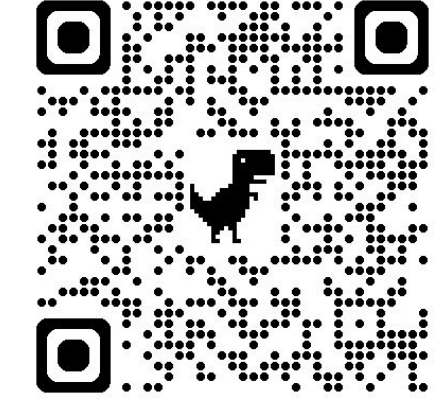


Code



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Motivation

CLT-based circuit discovery dominates

- Current query-level circuit discovery methods all build on surrogates like CLTs.
- CLT-based circuits place all nodes and edges on CLTs, requiring a strong assumption that they faithfully restore model computations.
- Question raised: Can we identify query-level circuits that reside directly within the original model?

Our Goal

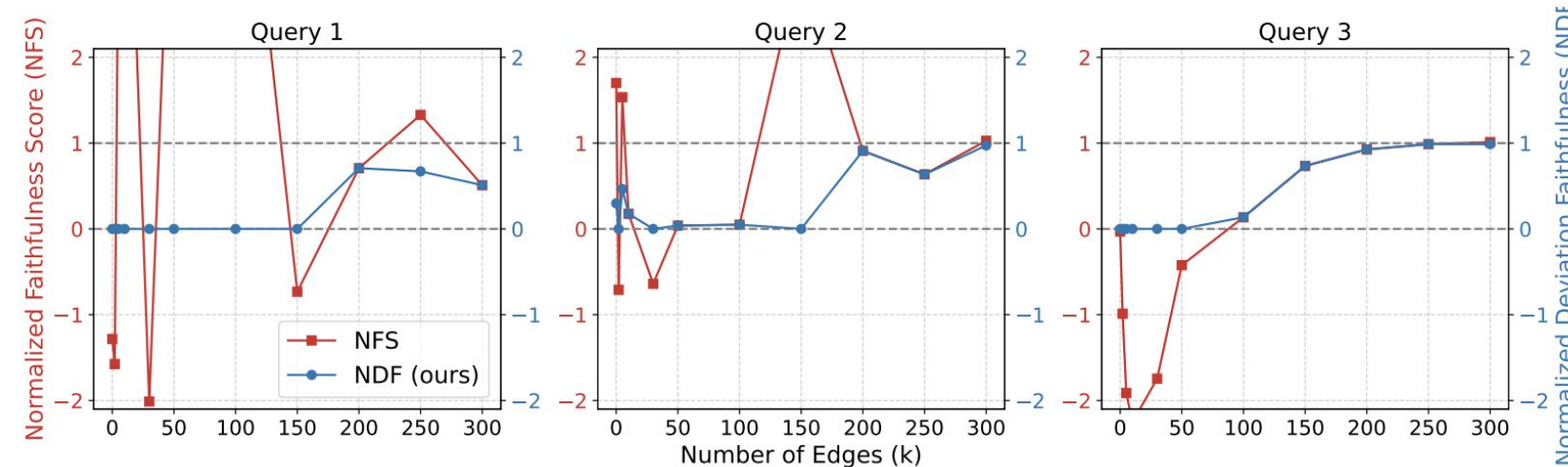
Query-level circuit discovery within model

- Explore the possibilities of query-level circuits within the original model.
- We argue: gradient-attribution with SAEs can be a potential alternative to CLTs.

Method 1 - NDF Score

The original NFS score to measure the circuit faithfulness is unreliable. We proposed the NDF score as an alternative:

$$NDF(C_q) = 1 - \min\left(\left|\frac{L(M(q)) - L(C_q(q))}{L(M(q)) - L(M(q'))}\right|, 1\right)$$

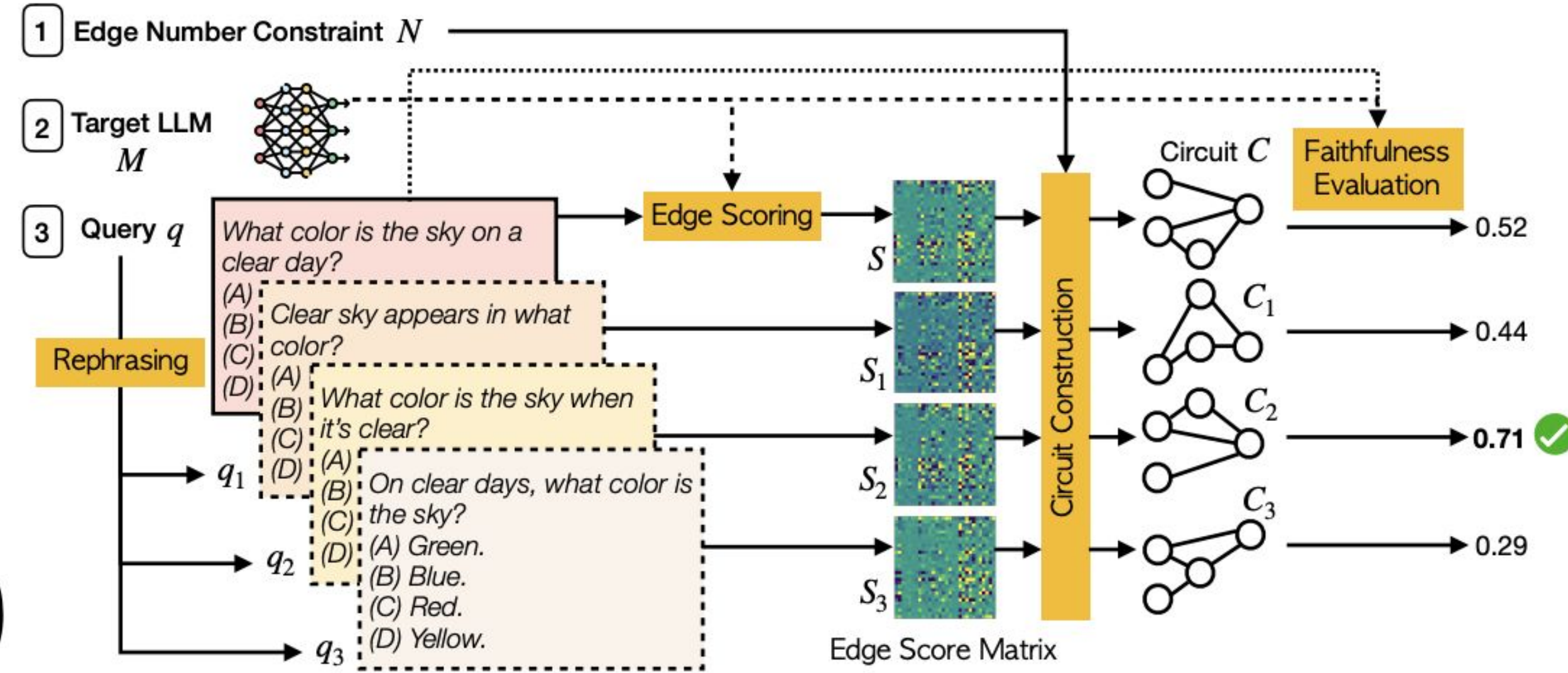


Method 2 - BoN on EAP-IG

We argue that BoN sampling via query paraphrasing, combined with integrated gradients (EAP-IG) for edge selection, enables the discovery of faithful query-level circuits within the original model.

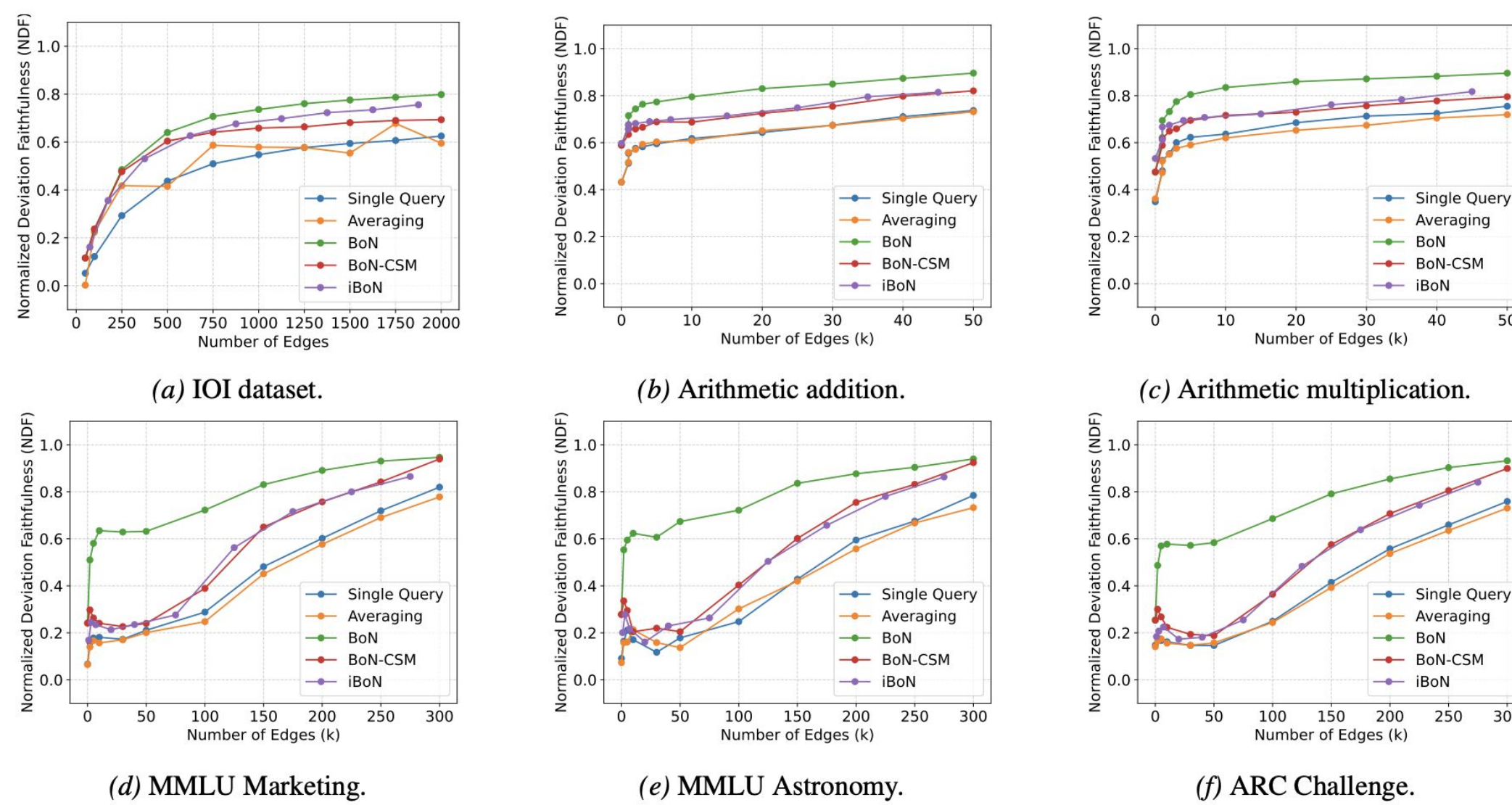
$$a_e = (e - e')^\top \int_0^1 \nabla_e M(z' + \alpha(z - z')) d\alpha$$

$$\approx (e - e')^\top \frac{1}{m} \sum_{k=1}^m \nabla_e M\left(z' + \frac{k}{m}(z - z')\right)$$

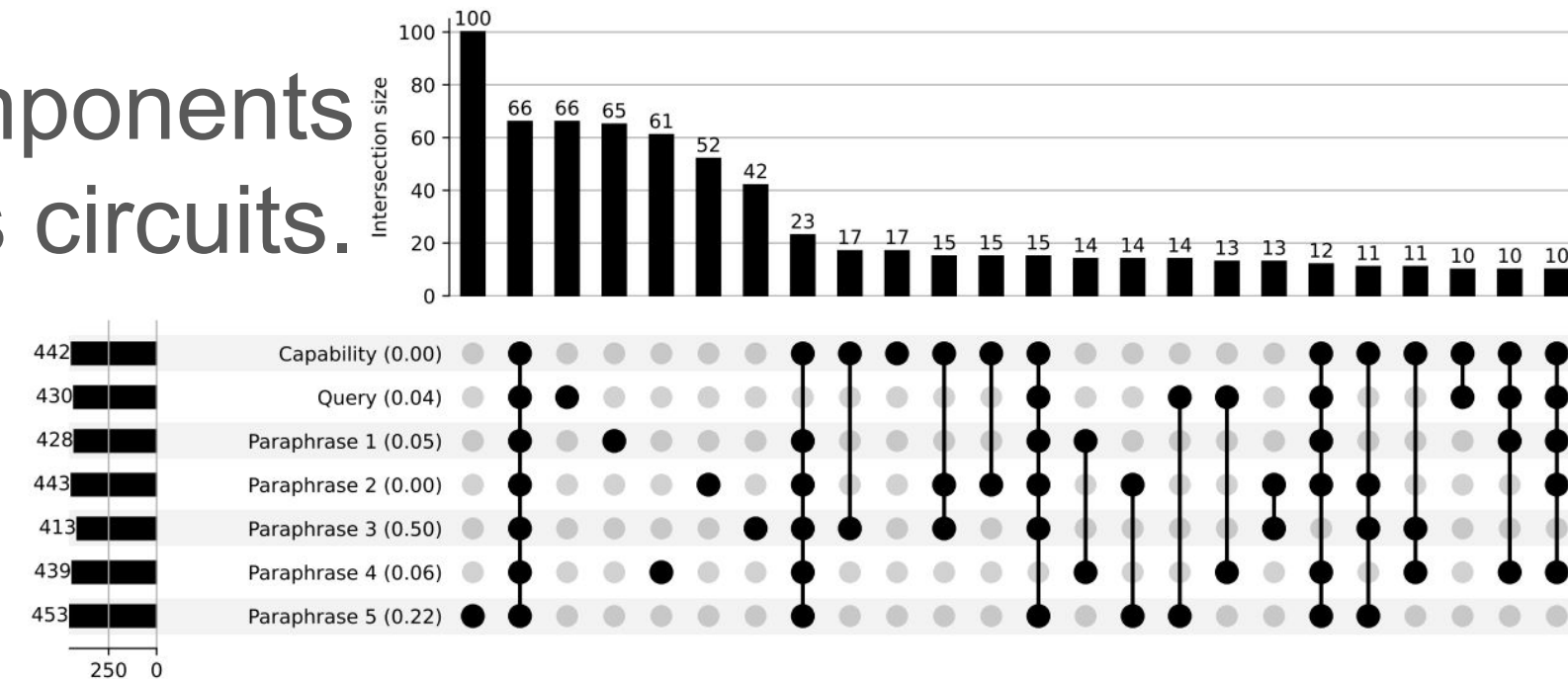


Results

BoN helps discover circuits with high NDF scores.



Shared components exist across circuits.



High-NDF query circuits robustly reflect more underlying model computations.

Deactivating gender-related SAE features within the discovered high-NDF circuits reduces the model's gender bias significantly more than low-NDF ones.

Metric	Scale	Circuit	Mean ± Std	W	p-value	Rosenthal's r
Absolute Bias Reduction	Logit	Best	0.810 ± 0.581	16.0	<0.0001	0.787
		Worst	0.234 ± 0.278			
	Δ Mean = +0.576		(****)			
	Probability	Best	0.063 ± 0.057	41.0	<0.0001	0.737
		Worst	0.011 ± 0.020			
	Δ Mean = +0.052		(****)			
Avg. Bias Reduction per Gender Feature	Logit	Best	0.073 ± 0.055	11.0	<0.0001	0.836
		Worst	0.014 ± 0.017			
	Δ Mean = +0.059		(****)			
	Probability	Best	0.006 ± 0.005	21.0	<0.0001	0.803
		Worst	0.001 ± 0.001			
	Δ Mean = +0.005		(****)			